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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning
SLOT : A

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

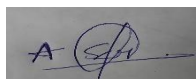
Course Outcome Weightage: 10%
Assessment Tool Weightage: 40%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I A. Santhosh Kumar Reddy/192525269(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Name of the student: KOTHAKOTA RAKESH

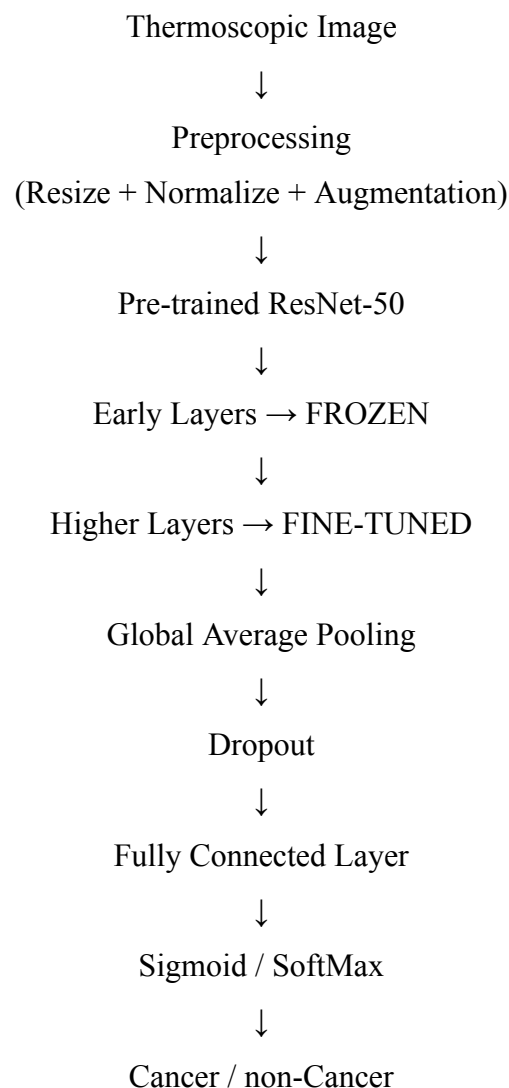
Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Q1. TRANSFER LEARNING

A HEALTHCARE STARTUP AIMS TO DEVELOP AN AI-BASED SYSTEM FOR DETECTING SKIN CANCER FROM DERMOSCOPIC IMAGES. SINCE ONLY A LIMITED NUMBER OF LABELED IMAGES ARE AVAILABLE, DESIGN A TRANSFER LEARNING STRATEGY USING A PRE-TRAINED CNN MODEL. JUSTIFY YOUR CHOICE OF MODEL, IDENTIFY WHICH LAYERS SHOULD BE FROZEN OR FINE-TUNED, AND EXPLAIN HOW YOUR DESIGN IMPROVES CLASSIFICATION PERFORMANCE.

Transfer learning is a deep learning technique in which a model that has already been trained on a large dataset is reused for a new but related task. In this case, the healthcare startup has only a limited number of labeled dermoscopic images. Training a CNN completely from scratch would require a large dataset and may cause overfitting. Therefore, a **pre-trained CNN such as ResNet-50** can be used.

Proposed Architecture



Component	Proposed Design
Pre-trained model	ResNet-50
Input	Thermoscopic image
Initial layers	Frozen
Higher CNN layers	Fine-tuned
Classifier	Fully connected layer
Activation	Sigmoid for binary classification
Loss	Binary Cross-Entropy
Optimizer	Adam
Augmentation	Rotation, flipping, scaling
Evaluation	Accuracy, Precision, Recall, AUC

The **initial layers should be frozen** because they learn general features that are useful for many image classification tasks. Freezing these layers also reduces the number of parameters that need to be trained and prevents overfitting when the medical dataset is small.

The **later convolutional layers should be fine-tuned** because they learn more

task-specific features.

Transfer learning reduces training time, requires fewer labeled images, and generally produces better accuracy than training a CNN from scratch. It also reduces the risk of overfitting and allows the system to learn useful visual features even with a limited medical dataset.

Conclusion:

A **ResNet-50 transfer learning model with frozen early layers and fine-tuned higher layers** is an effective solution for skin cancer detection. It provides efficient learning, better generalization, and improved classification performance with limited labeled data.

Q2. DENSENET AND PIXELNET

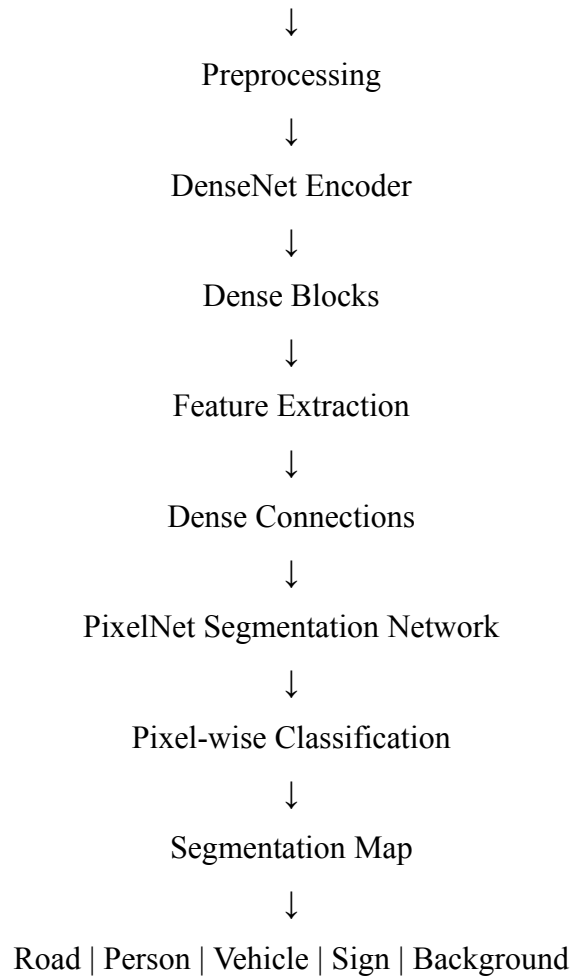
A SMART CITY PROJECT REQUIRES AN AUTOMATED ROAD-SCENE ANALYSIS SYSTEM THAT ACCURATELY IDENTIFIES ROADS, PEDESTRIANS, VEHICLES, AND TRAFFIC SIGNS AT THE PIXEL LEVEL. DESIGN A DEEP LEARNING ARCHITECTURE USING DENSENET AND PIXELNET TO SOLVE THIS PROBLEM. EXPLAIN HOW THE PROPOSED ARCHITECTURE IMPROVES FEATURE REUSE, SEGMENTATION ACCURACY, AND COMPUTATIONAL EFFICIENCY.

A smart city road-scene analysis system needs to identify **roads, pedestrians, vehicles, traffic signs, and other objects at pixel level**. This is a semantic segmentation problem because every pixel in the image must be assigned to an appropriate class.

A suitable architecture is a **DenseNet-based encoder combined with a PixelNet segmentation network**. DenseNet can be used for feature extraction, while PixelNet can generate pixel-level predictions.

Proposed Architecture

Road Scene Image



Component	Design
Input	RGB road image
Feature extractor	DenseNet
Segmentation network	PixelNet
Output	Pixel-level segmentation
Classes	Road, pedestrian, vehicle, sign, background
Activation	Softmax
Loss	Cross-Entropy / Dice Loss

Role of DenseNet

DenseNet connects each layer to all subsequent layers within a dense block. Instead of passing information only from one layer to the next, feature maps are reused throughout the network.

This provides **strong feature reuse**, improves gradient flow, and reduces the need to learn redundant features. DenseNet can therefore extract useful low-level and high-level features from complex road scenes.

For example, early layers can detect edges and shapes, while deeper layers can recognize vehicles, pedestrians, and traffic signs.

Role of PixelNet

PixelNet performs pixel-level classification using the features extracted by DenseNet. Instead of simply predicting one label for the complete image, it predicts a class for each pixel.

For example:

Pixel 1 → Road

Pixel 2 → Road

Pixel 3 → Vehicle

Pixel 4 → Pedestrian

Pixel 5 → Traffic Sign

Conclusion:

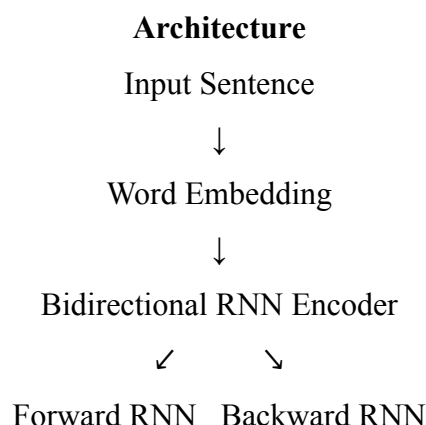
The combination of **DenseNet + PixelNet** provides a powerful road-scene segmentation system. DenseNet improves feature reuse and gradient propagation, while PixelNet performs pixel-level classification. The combined architecture can accurately identify roads, pedestrians, vehicles, and traffic signs while maintaining computational efficiency.

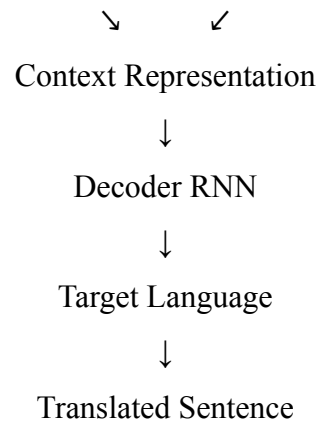
Q3. BIDIRECTIONAL RNN AND ENCODER–DECODER

A MULTINATIONAL CUSTOMER SERVICE COMPANY WANTS TO BUILD A MULTILINGUAL CHATBOT CAPABLE OF TRANSLATING CUSTOMER QUERIES WHILE PRESERVING CONTEXTUAL MEANING. DESIGN AN ENCODER–DECODER SEQUENCE-TO-SEQUENCE MODEL USING BIDIRECTIONAL RNNs. ILLUSTRATE THE

A multilingual chatbot needs to understand customer queries and translate them into another language while preserving the meaning and context of the sentence. An **Encoder–Decoder sequence-to-sequence architecture using Bidirectional RNNs** is suitable for this problem.

The encoder receives the input sentence one word at a time and converts it into a meaningful representation. A Bidirectional RNN processes the sentence in both forward and backward directions. Therefore, each word can be understood using information from both its previous and following words.





Component	Function
Input	Customer query
Embedding	Converts words into vectors
Encoder	Bidirectional RNN
Context	Represents complete sentence
Decoder	Generates translated sentence
Attention	Focuses on important input words
Output	Translated sentence
Loss	Cross-Entropy

A normal RNN processes the sentence only from left to right. This can make it difficult to understand information that appears later in the sentence. A **Bidirectional RNN** solves this problem by using two RNNs.

The forward RNN processes:

Word1 → Word2 → Word3 → Word4

The backward RNN processes:

Word4 → Word3 → Word2 → Word1

The outputs from both directions are combined to create a richer representation.

This architecture is particularly useful for multilingual translation because the same encoder-decoder framework can be trained for multiple language pairs.

Conclusion:

A **Bidirectional RNN Encoder–Decoder with attention** provides better contextual understanding than a simple unidirectional RNN. By considering both past and future information, it improves translation accuracy and helps preserve the meaning of multilingual customer queries.

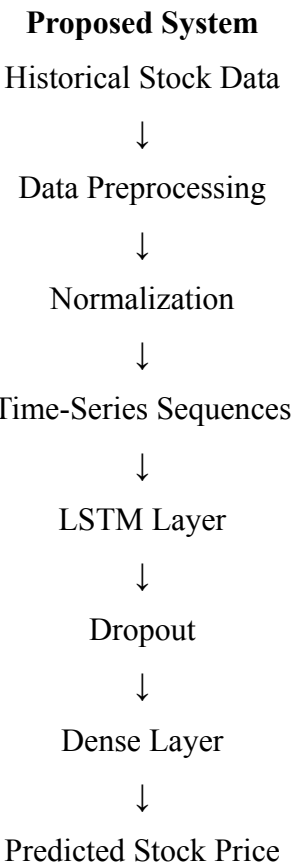
Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to

train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Stock prices depend on historical patterns and relationships over time. Therefore, a recurrent neural network architecture is suitable for processing stock market time-series data. However, traditional RNNs can suffer from the **vanishing gradient problem**, making it difficult to learn long-term dependencies.

An **LSTM (Long Short-Term Memory)** network is more suitable because it contains memory cells and gates that control the flow of information.



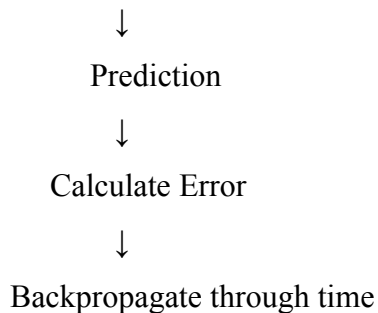
Component	Design
Input	Historical stock prices
Features	Open, High, Low, Close, Volume
Preprocessing	Normalization
Model	LSTM
Regularization	Dropout
Output	Future stock price
Loss	Mean Squared Error
Optimizer	Adam
Metrics	MAE, RMSE

Role of BPTT

Backpropagation Through Time (BPTT) is used to train recurrent networks. During training, the sequence is unfolded across time steps.

For example:

$t1 \rightarrow \text{LSTM} \rightarrow t2 \rightarrow \text{LSTM} \rightarrow t3 \rightarrow \text{LSTM} \rightarrow t4$



Forget gate – decides which old information should be removed.

Input gate – decides which new information should be stored.

Output gate – decides which information should be used for prediction.

Traditional RNN	LSTM
Simple memory structure	Memory cell with gates
Suffers from vanishing gradients	Handles long-term dependencies better
Difficult to remember old information	Can retain important information
Less suitable for long sequences	Suitable for time-series data

Stock prices may contain patterns that depend on information from many previous time steps. LSTM can preserve important historical information and ignore irrelevant information.

However, stock prices are affected by many unpredictable factors such as economic events, company announcements, and market sentiment. Therefore, LSTM predictions are estimates and cannot guarantee future prices.

Conclusion:

An **LSTM trained using BPTT** is suitable for stock-price forecasting because it can learn temporal dependencies and retain important historical information. Compared with a traditional RNN, LSTM provides better handling of long-term dependencies and more stable training.

Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer

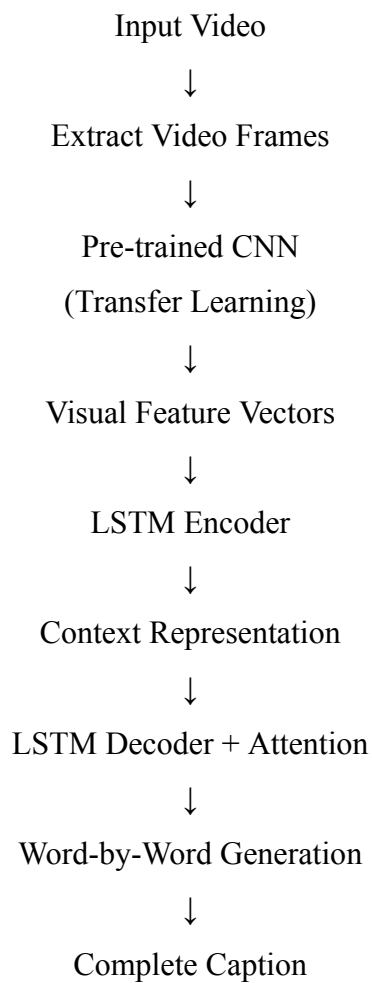
Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for caption generation. Explain the role of each component and justify how the complete system produces accurate

captions.

Automatic video captioning combines **computer vision and natural language processing**. The system must first understand the visual content of a video and then generate a meaningful sentence describing that content.

A suitable solution combines **Transfer Learning for visual feature extraction** with an **LSTM Encoder–Decoder** for caption generation.

Proposed Architecture



Component	Function
Input	Video
Frame extraction	Converts video into frames
CNN	Extracts visual features
Transfer Learning	Uses pre-trained CNN knowledge
LSTM Encoder	Learns temporal information
LSTM Decoder	Generates caption
Attention	Focuses on important frames/features
Output	Natural-language caption

Component	Function
Loss	Cross-Entropy

Transfer Learning Component

Training a CNN from scratch for video understanding requires a very large dataset. Instead, a CNN such as **ResNet, EfficientNet, or another pre-trained image model** can be used.

The pre-trained CNN extracts useful features such as objects, shapes, colors, and scene information from each video frame.

For example:

Frame 1 → CNN → Feature Vector 1

Frame 2 → CNN → Feature Vector 2

Frame 3 → CNN → Feature Vector 3

Frame 4 → CNN → Feature Vector 4

These feature vectors are then passed to the LSTM.

LSTM Encoder–Decoder

The **LSTM encoder** receives the sequence of visual features and learns the temporal information in the video. It understands how the scene changes from one frame to another.

The **LSTM decoder** generates the caption one word at a time.

An attention mechanism can help the decoder focus on relevant video frames when generating each word. For example, while generating the word "**cricket**", the model can focus more strongly on frames showing the cricket activity.

Transfer learning reduces the amount of training data required because the CNN already contains useful visual knowledge. LSTM handles the temporal sequence, while the decoder converts the learned representation into natural language.

Conclusion:

The combination of **Transfer Learning + CNN + LSTM Encoder–Decoder + Attention** provides an effective video-captioning system. Transfer learning extracts strong visual features, the LSTM encoder understands temporal information, and the LSTM decoder generates meaningful captions word by word. This integrated approach is suitable for accessibility applications because it can automatically describe video content for users who cannot fully access the visual information.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understandin g of Industry Problem	20%	Comprehensiv e understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts

		solve the problem			
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis